Vandan Patel

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EDUCATION

Rutgers University–New Brunswick | Honors College — **New Brunswick, NJ**
B.S. Computer Science, B.S. Finance, B.A. Data Science — *May 2027*

- **GPA:** 3.96/4.00 | **Honors & Awards:** Phi Beta Kappa, Dean's Scholarship, Dean's List (All Semesters)
- **Coursework:** Data Structures, Algorithms, Computer Architecture, Intro to AI, Computers in Biomedicine (Grad)

EXPERIENCE

Amazon — **Bellevue, WA**
Financial Analyst Intern — *May 2026 – Aug 2026*

- Built an AI agent harness using the Strands Agents SDK and Amazon Bedrock; deployed on EC2 and adopted by 30+ finance employees for cross-team knowledge discovery and resource ownership lookup, reducing meeting overhead
- Developed a 15-tool MCP server enabling AI agents to query and update a Python knowledge graph spanning 15+ teams, with authorized writes and Git-backed change tracking
- Engineered a Python memory extraction pipeline that identified organizational facts from Markdown-based agent memories, using human-in-the-loop approval and SSO-enforced writes to update a shared knowledge graph
- Implemented a SQL attribution framework linking 24 months of automated inventory recommendations to actual liquidation shipments, uncovering 31% of total costs outside the standard recommendation process

Rutgers SEBS, Infrastructure & Facilities Planning — **New Brunswick, NJ**
Data Analyst Intern — *Sep 2025 – Dec 2025*

- Enabled semiannual audits for 300+ buildings by automating record checks and updates, saving 40+ hours annually
- Cut verification cycles by ~35% with centralized submission tracking and automated follow-ups for missing responses

Moweb Technologies — **Secaucus, NJ**
Software Engineering Intern — *Jul 2025 – Aug 2025*

- Reduced weekly ETL pipeline failures from 20+ to ~1 by resolving NULL-handling issues in dbt SQL transformations
- Collaborated on building dbt reporting tables during a MySQL-to-Snowflake migration, cutting query times by 40%

PROJECTS

LLM Fine-Tuning & Inference Evaluation | *Python, MLX, Qwen3, LoRA* — *Sep 2026 – Present*

- Built Python/MLX pipeline to test when LoRA fine-tuning could cost less than repeatedly supplying task instructions
- Fine-tuned Qwen3 8B; rejected faster setup after scoring 79/96 on synthetic validation versus 96/96 for the baseline
- Reduced mean Qwen3 8B latency by 18% (6.99s to 5.73s) via KV-prefix caching, preserving validation accuracy

GALE — Algorithm Practice Platform | *JavaScript, Gemini API* — *Apr 2026 – Present*

- Launched an algorithm-practice PWA with 20 patterns, mastery progression, weakness targeting, and spaced repetition
- Engineered Gemini grading with rule-based checks, JSON validation, three-model failover, and self-grading fallback
- Implemented AI-reviewed grading appeals by restoring pre-grading state and reapplying corrected grades to reconcile mastery, Elo ratings, review schedules, and session results without double-counting answers or streak credit

GrindSheet — iOS Workout Tracker | *SwiftUI, Supabase, HealthKit* — *Sep 2025 – Present*

- Developed a SwiftUI workout app with Supabase authentication, synced exercise logs, and HealthKit activity tracking
- Implemented workout-note parsing and fuzzy exercise matching, flagging uncertain matches for review before saving
- Computed 7- and 28-day training loads from HealthKit and exercise logs, deduplicating overlapping strength sessions

LEADERSHIP AND COMPETITIONS

Rutgers Shark Tank — **Piscataway, NJ**
Technical Lead, PerkPal — *Mar 2025*

- Collaborated with a 3-person team to align product strategy, business case, and live demo, finishing top 6 of 30+ teams
- Led development of a React/Selenium rewards-aggregation platform and presented the prototype to an investor panel

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, Swift
Frameworks & Libraries: FastAPI, Flask, React, SwiftUI, dbt-core, MLX, Strands Agents SDK
Data, Infrastructure & Tools: PostgreSQL, Snowflake, Supabase, Docker, Git, AWS (EC2, S3, Bedrock), MCP